Zander Schatzberg

Cornell University: Computer Science, Mathematics, and Cognitive Science
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Relevant Experience

Musharoff Lab (Computational Genomics), Cornell University Jan 2026 – Present
Undergraduate Researcher, advised by Prof. Shaila Musharoff *Ithaca, NY*

- Computing polygenic risk scores (PRS) for breast cancer and comparing their predictive accuracy across phenotypic strata.
- Identifying polymorphic human endogenous retrovirus (HERV) insertions and examining underlying population structure.
- Testing gene-by-environment (GxE) interactions in canines to measure how various chemical exposures may modulate oncological and behavioral phenotypes
- Building genetic relationship matrices and PRS calculations in support of other lab projects.

Cornell ALPS Lab (High-Performance Computing) Jan 2025 – Present
Undergraduate Researcher, advised by Prof. Giulia Guidi *Ithaca, NY*

- Developed a kernel for sparse matrix multiplication on processing-in-memory hardware.
- Accelerating GxE interaction testing using a novel application of genotype representation graphs.

Rekall.ai Summer 2025
Data Engineering Intern *New York, NY and Amsterdam, Netherlands*

- Improved multi-view 3D ball tracking system using both classical computer vision algorithms and deep learning, and generated simulated data to train the tracking model.
- Improved the accuracy of delivery time estimates for Teknova (a chemical reagent supplier), producing a more consistent range of expected delivery dates.

UCSC Department of Astrophysics, Thacher Observatory Fall 2022 – Summer 2024
Research Intern *Ojai, CA*

- Developed a program that autonomously searches for merging neutron stars and black holes detected by gravitational wave observatories, allowing our observatory to contribute to frontier research on the origins of heavy elements.
- Developed a dashboard to coordinate transient follow-up across UCSC, Northwestern, and Thacher research groups, decreasing latency between discovery and observation for several object classes.

Manuscripts

- Socioeconomic status affects breast cancer PRS performance. Co-1st author, under internal review. *(Musharoff Lab)*
- Polymorphic HERVs and their underlying population structure. 1st author, in preparation. *(Musharoff Lab)*
- Accelerating GxE interaction testing with genotype representation graphs. 2nd author, in preparation. *(ALPS Lab)*

Notable Projects

Robotic Studio July 2026 – Present
Web platform that accelerates robot design by integrating 3D CAD, PCB design, simulation, part sourcing, and assembly instructions behind an interface that combines natural language with visualization.

Snorfel Dec 2024 – July 2026
iOS app providing a social forum for pet owners. Cofounder and engineer.

Skills + Interests

General:	High-performance computing, numerical linear algebra, statistical genetics/genomics, computer vision
Languages:	Python, C, TypeScript/JavaScript, Java, OCaml, SQL, RISC-V Assembly
ML & Data:	PyTorch, TensorFlow, NumPy, pandas, SciPy, OpenCV
Web & Infra:	SvelteKit, Node, PostgreSQL, Docker, Git, AWS, Google Cloud, Modal, Railway, Vercel, Cloudflare
Interests:	Ultramarathons, reading, cooking, bow hunting, and jiu jitsu

Education

Cornell University Aug 2024 – May 2028
Bachelor of Arts: Majors in Computer Science and Mathematics, minor in Cognitive Science *Ithaca, NY*

- **Relevant coursework:** Object-Oriented Design and Data Structures (Honors), Machine Learning, Discrete Math, Functional Programming, Computer System Organization, Operating Systems, Analysis of Algorithms, Natural Language Processing, Linear Algebra, Multivariable Calculus, Computational Neuroscience, Number Theory, Neurobiology
- Pursuing an honors thesis in Computational Biology.